

and other sensors and meters. By using it, you can easily manage your home devices with your mobile phone. Domoticz can be installed and used on Raspberry Pi. Let us take a look at how this is done.

What you need

- Power supply and source
- MicroSD card
- Raspberry Pi
- Ethernet cable
- Microphone



Note: Set up the Raspberry on a static IP address.

- i. Upgrade all the packages on the Raspberry Pi by running the following commands:

```
sudo apt update
sudo apt upgrade
```

- ii. Install Domoticz in the Raspberry Pi by running the install script from its website. Run the following command:

```
curl -L https://install.domoticz.com
| bash
```

After the script run completes, press the *Enter* key.

You will then choose the services you want Domoticz to operate on. For this tutorial, we will leave both HTTP and HTTPS on. But some people prefer to disable the HTTP so that they can use the more secure HTTPS. Press *Enter* to continue.

Choose the port you want HTTP to run on, although this is usually by default 8080; so just press *Enter*. And do the same for the HTTPS port, which is 443 by default.

- iii. Choose the Domoticz installation folder, which usually by default is the *home* folder. Press *Enter*.
- iv. The installation process is complete at this point and you will be shown two IP addresses, which you

should take note of. Press *Enter* to complete the installation process.

Configuration of the Domoticz

Web interface

Get the static IP address we assigned to the Raspberry Pi above. Through a Web browser, run:

```
http://[IPADDRESS]:8080
```

At *IPADDRESS* use the static Raspberry IP address.

At this point you will have access to the Domoticz Web interface and can now monitor your various connected home appliances.

Raspberry Pi Bluetooth speakers

This project is a little bit more difficult if you do not have the right software tools; it is for this reason that we will use the Balena image (*balenaSound*) which will simplify our work. This can enable audio from Bluetooth and Spotify, as well as other synchronised audio.

What you need

- Power supply and source
- MicroSD card
- Raspberry Pi
- Ethernet cable
- Bluetooth adapter if it is not already available

Setup steps

Download *balenaSound* image for use with Bluetooth. You have to have a Balena account to proceed.

- i. Create an application with the Balena Web interface.
- ii. On the screen that appears, choose the *Default Device Type* and select the Raspberry Pi that you will use as a default device. After that, click *Create and deploy*.
- iii. On the page you are taken to with the devices you have deployed, click on *Add device* to add Bluetooth speaker software, as it is not on the list.

- iv. A pop-up that will allow you to create a balenaOS image for the device you added in (iii) above appears. Select the device type; then choose the version of the balenaOS image to use. For this purpose, we choose the *Production* option. Another option is the *Development* option, but this is used for troubleshooting purposes.

To use Wi-Fi, click the toggle under *Network Connection*; but you must also specify its SSID and passphrase.

- v. Now download balenaOS.

Adding the balenaOS to your Raspberry Pi

To add the balenaOS to your Raspberry Pi, we will use the Etcher tool to write the *balenaSound* image to the Raspberry Pi's SD card.

- i. In the Etcher software, click on the *Select Image* option. From the file dialogue that appears, select *balenaSound* image to write to the Raspberry SD card.
- ii. Choose the image you want to write the image to.
- iii. Click on *Flash* to write it to your SD card. Note that the drive you choose will be wiped.

After completion, you can now plug the drive into the Raspberry Pi to run.

Checking the status of your Raspberry Bluetooth speakers on Balena

- i. Go to the Balena dashboard page and click the *balenaSound* application.
- ii. A list showing the devices running on this application will appear, and from it you can check the status of yours.
- iii. There is a *services* section from which you can see the Bluetooth audio service, Spotify and Airplay. From the interface, you can control the services by taking actions such as *start*, *restart* or *stop*.